

(d), at a V_{DS} of 100 mV, the output resistance is only 25k. Using the transconductances in (e) and (f), the open circuit gains are, roughly, 4.375 (NMOS) and 3.125 (PMOS). In the actual circuits, the biasing points will vary (but in any case the gain of single stages will be low).

The Differential Amplifier

Figure 26.19 shows a cascode load diff-amp based on the topology seen in Fig. 26.7. Seen in the figure are typical values for the voltages in the circuit assuming gate-source voltages of 400 mV and, for the bottom two rows of NMOS devices, drain-source voltages of 100 mV. Notice how we used V_{biasn} to bias the second row of PMOS devices, which puts 200 mV across the drain-source voltages of the PMOS devices. This results in larger gain and reduced output swing. Diff-amp output swing is not an issue if this amplifier is used as the first-stage in a two-stage op-amp.

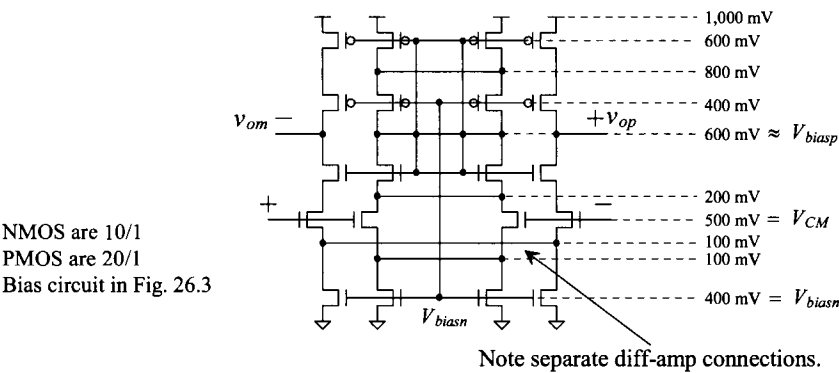


Figure 26.19 Fully-differential cascode diff-amp.

Figure 26.20 shows how the output voltages of the diff-amp in Fig. 26.19 vary with changes in V_{CM} . A 200 mV change in the common-mode voltage results in a, roughly, 50 mV change in the diff-amp's common-mode output voltages. Looking at Fig. 26.1, we see that the change in the drain current will be, again roughly, $\pm 5 \mu A$ around the quiescent value set with V_{CM} equal to 500 mV.

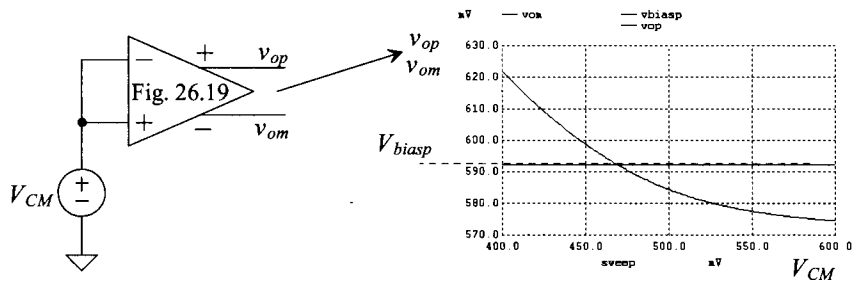


Figure 26.20 Varying the common-mode voltage and looking at the output.

Figure 26.21 shows the DC characteristics of the diff-amp. The gain is approximately 40. An important concern is how the mismatches in the MOSFETs used in the diff-amp affect the operation and biasing of the amplifier. Before discussing this issue, let's add the second stage and CMFB circuitry to form an op-amp. Note that, with a diff-amp gain of 40, and a second stage gain of 10, our op-amp's open loop gain will only have a value in the hundreds.

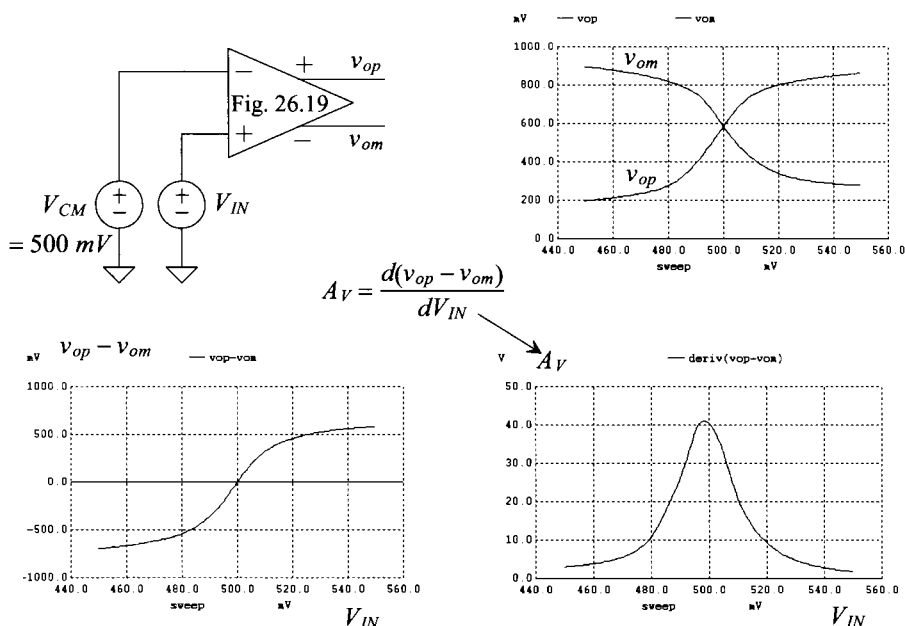


Figure 26.21 DC behavior and gain of the diff-amp in Fig. 26.19.

Adding a Second Stage (Making an Op-Amp)

Figure 26.22 shows a two-stage op-amp without CMFB circuit. The second stage of the op-amp operates class AB, as seen in Fig. 26.2, and the associated discussion. We've spent a considerable amount of time discussing how the output voltages of the diff-amp are approximately V_{biasp} . It should be clear after studying the op-amp in Fig. 26.22 why this is important. This voltage sets the quiescent current flowing in the output stages. There are eight vertical branches in this op-amp, so we can estimate the current pulled from V_{DD} under quiescent conditions as $160 \mu\text{A}$.

We used 50 fF capacitors for compensation in this op-amp. We can estimate the slew-rate limitations caused by the diff-amp driving the compensation capacitor, Eq. (22.34), as $20 \mu\text{A}/50 \text{ fF} = 400 \text{ mV/ns}$. Using a class AB output stage, we don't have slew-rate limitations associated with driving a load capacitance from a constant current source. As discussed at the beginning of the chapter, the output MOSFET's drain currents can be pulsed to a value greater than $100 \mu\text{A}$. When the op-amp is driving a 250 fF capacitive load, the speed limitations associated with charging the load capacitance are similar to the limitations we get when the diff-amp drives the compensation capacitor.

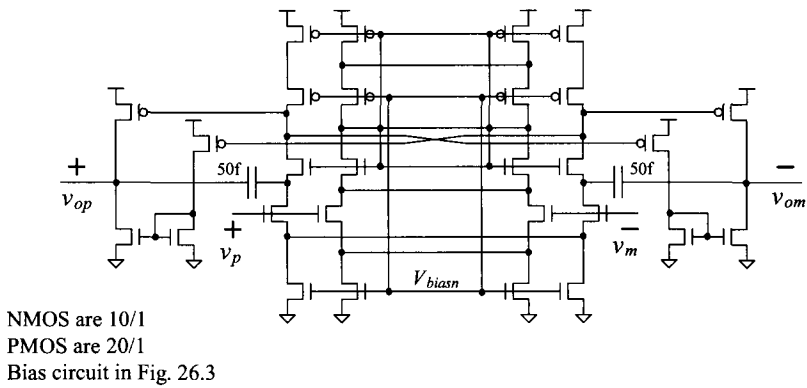


Figure 26.22 Basic two-stage op-amp without CMFB.

Figure 26.23 shows the DC characteristics of the op-amp in Fig. 26.22 where, once again, we've held the inverting op-amp input at the common-mode voltage and swept the voltage on the noninverting input. Notice how the outputs swing all the way from ground to V_{DD} ($= 1\text{V}$). Further notice how the differential output voltage swings from -1 to $+1\text{V}$ (a doubling in the output swing as discussed in Sec. 25.2.) The op-amp's DC gain is approximately 500 without a DC load. Notice how the two op-amp outputs cross at 400 mV (not at V_{CM} where they should). We'll discuss the CMFB in a moment.

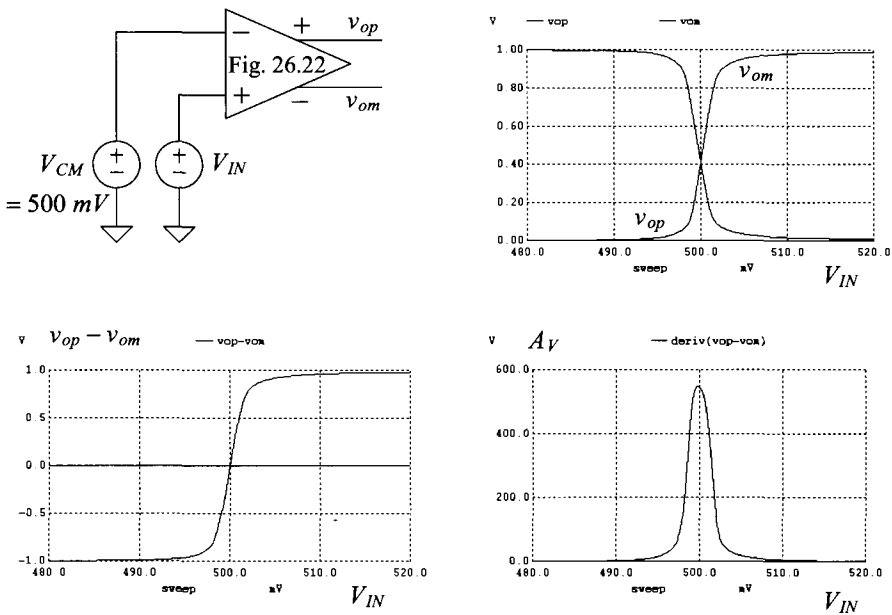


Figure 26.23 DC behavior and gain of the op-amp in Fig. 26.22.

Step Response

To determine both the stability of the op-amp in Fig. 26.22 and the settling time under certain loading conditions, consider the configuration seen in Fig. 26.24. We used relatively small resistors, 20k, in this circuit to reduce the RC time constant associated with the outputs of the op-amp charging the input capacitance of the op-amp. If, for example, the input capacitance of the op-amp is 25 fF (from parasitics and the MOSFETs used on the op-amp's input), then the RC time associated with charging this capacitance through a 20k resistor is 0.5 ns. As seen in Fig. 26.24, the settling time is approximately 2.5 ns so this RC time can have a significant effect on the settling time and stability of the circuit (important).

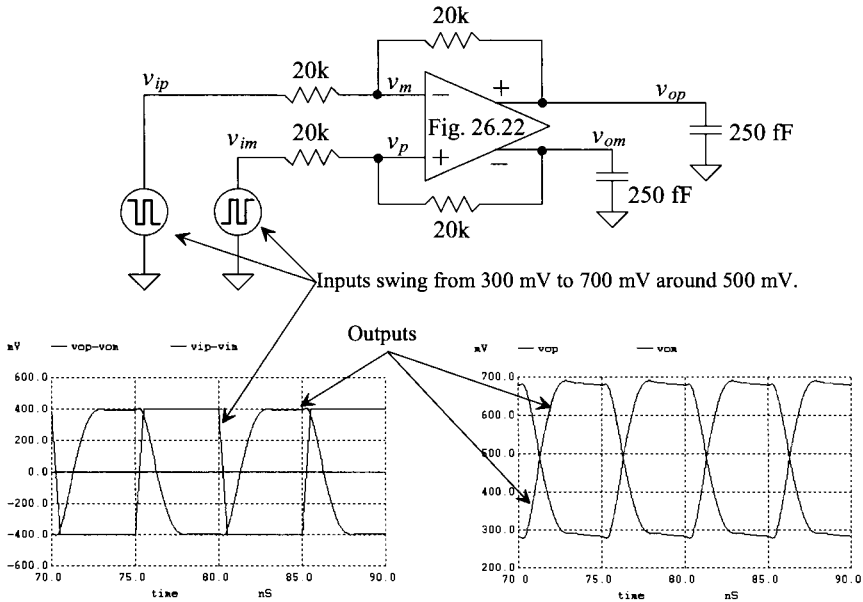


Figure 26.24 Step response of the op-amp in Fig. 26.22 driving 250 fF load capacitors and 20k feedback resistors.

Notice how, in Fig. 26.24, we used input signals that don't swing rail-to-rail. Since we don't have a CMFB circuit in the op-amp, the exact common-mode output voltage is an unknown. It may be 400 mV or it may be 600 mV. We get some help in setting the circuit's output common-mode level by using input signals with common-mode voltages of 500 mV and DC feedback. Looking at the simulation results in Fig. 26.24, we might get a false sense of not needing a CMFB circuit. To illustrate this is indeed a false sense, consider the sample-and-hold seen in Fig. 26.25 (see also Fig. 25.19). When the ϕ switches are closed, the op-amp's inputs and outputs should be held to $V_{CM} \pm V_{OS}$ (the op-amp is placed in the follower configuration). As seen in the figure, the outputs during this time are driven close to 400 mV. On closer inspection, we see the output common-mode level is wandering downwards until eventually the op-amp shuts off.

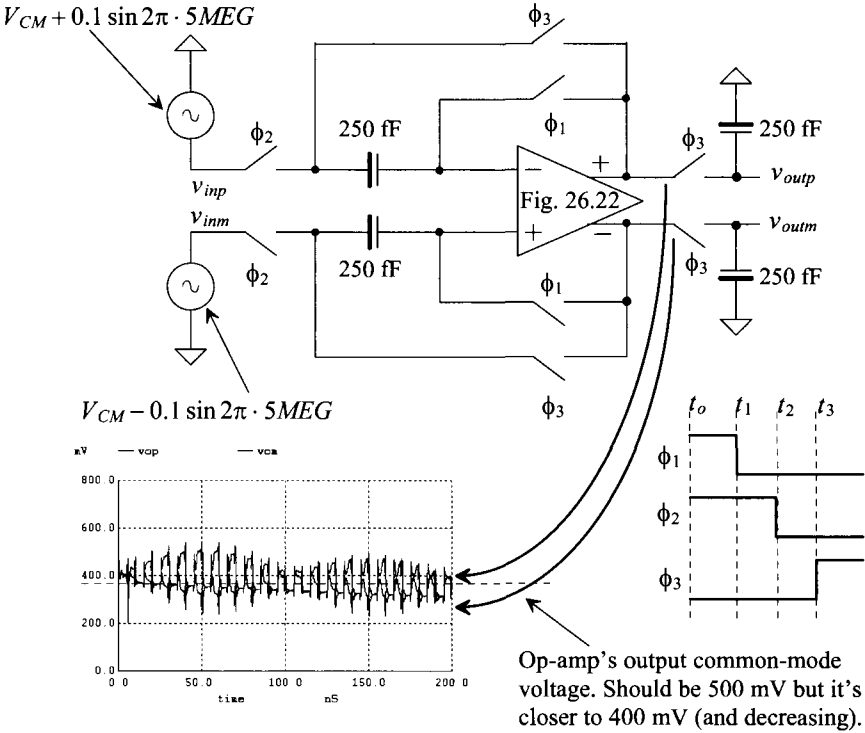


Figure 26.25 A sample-and-hold circuit. Notice how the output common-mode voltage is wandering.

Adding CMFB

Figure 26.26 shows how we can modify the basic op-amp to allow for a CMFB signal input. If the outputs of the op-amp (their average or common-mode voltage) are too high,

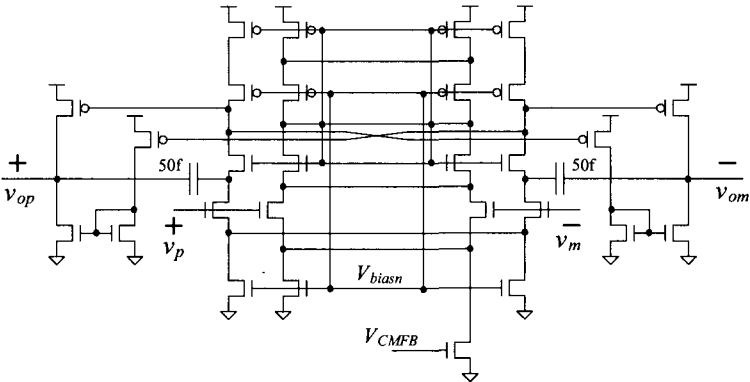


Figure 26.26 Modifying the op-amp for a CMFB input signal.

V_{CMFB} goes up. This causes the output voltage of the diff-amp to go up (increasing V_{CMFB} causes the bias current in the center MOSFETs of the diff-pair to increase). An increase in the diff-amp's output voltage lowers the quiescent current flowing in the output buffers, causing the output voltage to move downwards. This is a good time to remember one of the fundamentals from Ch. 20, namely, if two MOSFET gate-source voltages are equal and they have the same drain current, then their drain-source voltages must be equal. For the output buffer in Fig. 26.26, this means that as we reduce the drain currents flowing in the output buffer (by driving V_{CMFB} high), the gate-source voltages of the NMOS devices decrease. Because of the fundamental concept just mentioned, this causes the output voltages of the op-amp to decrease. Figure 26.27 shows the op-amp's output voltage change with V_{CMFB} . For stability concerns, it's of interest to determine the gain from the CMFB input to the outputs. From the simulation, the gain is approximately 25 (about 10 times less than the differential gain). The lower forward CMFB gain allows us to use a diff-amp with a current mirror load for the CMFB amplifier (discussed next).

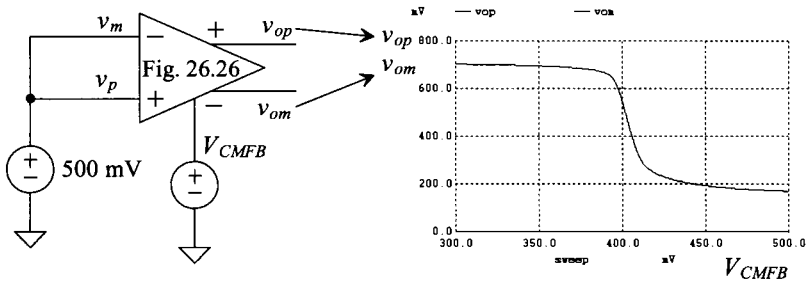


Figure 26.27 The CMFB input to output relationship. The gain is approximately 25 (considerably less than the forward differential gain).

CMFB Amplifier

Now that we know how our CMFB signal, V_{CMFB} , in the op-amp of Fig. 26.26 affects the output voltages and that the gain this signal sees is approximately 25 ($v_{op,n}/V_{CMFB}$), we need to discuss the CMFB amplifier. Consider the two diff-amps seen in Fig. 26.28. In the top diff-amp, gate-drain-connected loads are used. As seen in the simulations, the gain from the averaged input, V_{CMA} , to the diff-amp's output, V_{CMFB} is considerably less than 1 (and so the CMFB is guaranteed to be stable). Unfortunately, the CMFB amplifier's output voltage isn't high enough. (As seen in Fig. 26.27, we need approximately 400 mV.) The diff-amp's tail current can be increased in size (use at least two PMOS for the tail current) to increase the output voltage. However, the low gain, say around 0.3, combined with the CMFB gain through the op-amp, again around 25, means that the CMFB loop's overall gain is only around 7. This isn't large enough to precisely balance the outputs. Using the diff-amp with current mirror load, Fig. 26.28b, gives a gain of approximately 8. This combined with the op-amp's CMFB gain gives an overall CMFB loop gain of 200. This is less than the op-amp's differential gain (so we can use the same compensation capacitors to stabilize the CMFB loop), Fig. 26.23, and large enough to precisely balance the op-amp's outputs.

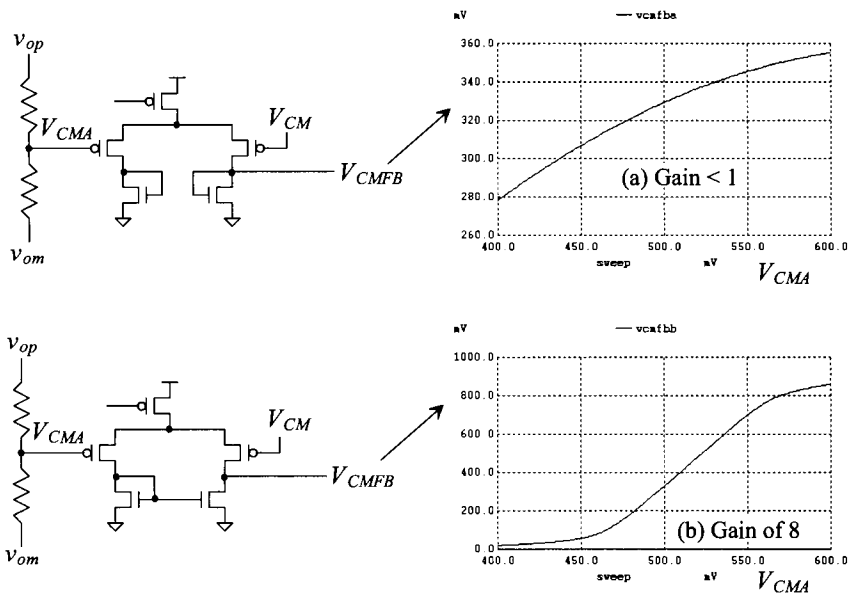


Figure 26.28 Gains of CMFB amplifiers.

The Two-Stage Op-Amp with CMFB

Figure 26.29 shows the complete schematic of the op-amp. Notice that we've doubled the width of the MOSFETs in the output buffer. The added loading from the CMFB averaging resistors will lower the open-loop gain. To compensate for this reduction, the

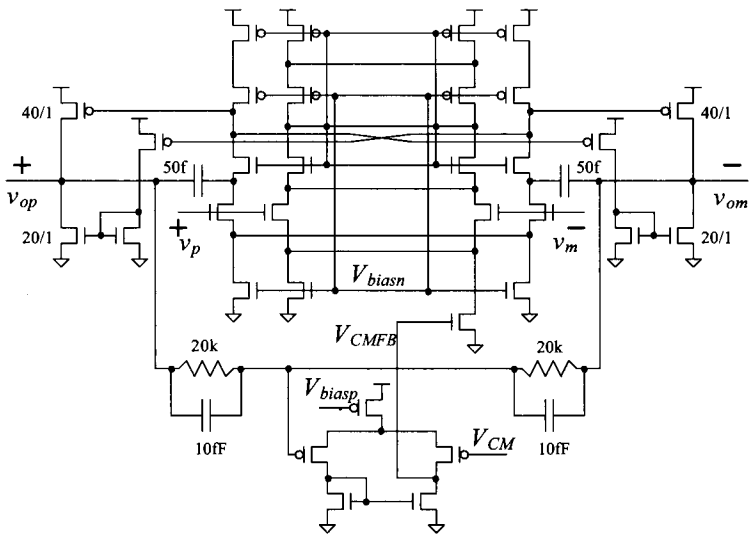


Figure 26.29 Complete schematic of op-amp with CMFB.

drive strength of the MOSFETs on the output of the op-amp was increased. Figure 26.30 shows the DC behavior of the op-amp when placed in the configuration seen in Fig. 26.23. Notice that the outputs cross at the ideal common-mode voltage of 500 mV. The gain is reduced because of the loading by the 20k resistors in the CMFB circuit (and so the gain can be increased by increasing the values of the resistors in the CMFB circuit). However, **we see a problem** with these simulation results. The outputs are only swinging from 200 to 800 mV (not from 0 to V_{DD} as possible with a push-pull output stage). Further, we can estimate the current pulled from V_{DD} for the op-amp in Fig. 26.29 by counting the number of branches in the op-amp (noting that the output stage counts twice because we've doubled the widths of these devices). Including the CMFB amplifier, there are 11 branches. If the bias current through each branch is 20 μA and the current pulled by the bias circuit is 50 μA , then we would expect the op-amp to pull 270 μA (perhaps even a little less because many of the MOSFETs are biased near the triode region). When we look at the simulation results that generated Fig. 26.30, we see that the current is closer to 500 μA (way off indicating, again, that we've got a problem).

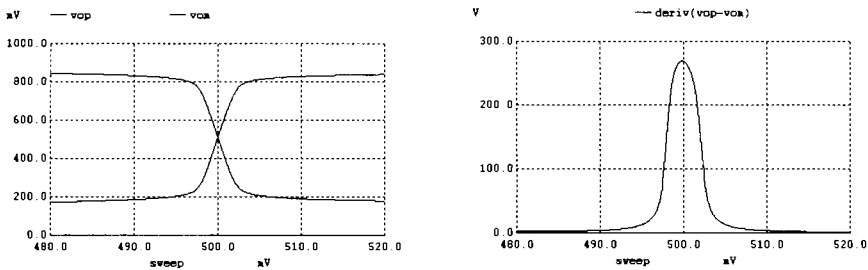


Figure 26.30 Simulating the operation of the op-amp in Fig. 26.29.

Origin of the Problem

The origin of the problem (that is, the op-amp drawing too much current and the output swing not reaching the rails) can be traced to the output buffer. We spent a considerable amount of time discussing how we want to set the outputs of the diff-amp to V_{biasp} . When we look at the simulations, we see the diff-amp's outputs are considerably less than the ideal 600 mV of V_{biasp} . To understand why, let's look at the output buffer in Fig. 26.31. Since the problems appear with the addition of the CMFB circuit, we assume that the two gates of the PMOS devices are moving at the same potential (tied together), that is, that the inputs to the buffer are moving with the diff-amp's output common-mode level. When the output of the diff-amp is V_{biasp} (600 mV), 20 μA flows in all of the MOSFETs. (To keep things simpler, we don't include the doubling in the widths used in the op-amp output devices.) The gate potentials of the NMOS are, roughly, V_{biasn} (400 mV). Because of the symmetry of the circuit, this means that the output is also at 400 mV. To increase the output voltage to 500 mV, we must drop the potential on the gates of the PMOS until the gate-drain-connected NMOS has a V_{GS} of 500 mV (again because of the symmetry). This increases the current (significantly) flowing in the output buffer, lowers its gain, and reduces the linear output swing. The overdrive voltages essentially change from the desired 100 mV when 20 μA of current flows to 200 mV when the output is driven to 500 mV.

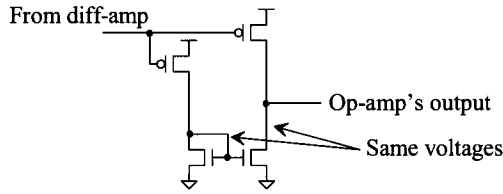


Figure 26.31 Output buffer used in the op-amp of Fig. 26.29.

Figure 26.32 shows one solution to this problem. We've added a device to cascode the output buffer's NMOS (the one connected to the output terminal). The speed shouldn't be affected by the addition of the device (which operates near or in the triode region). The added device allows the op-amp's output to swing more freely (but the current still won't be precisely set). The added device won't affect the output swing range of the op-amp. Figure 26.33 shows the op-amp with modified output buffer.

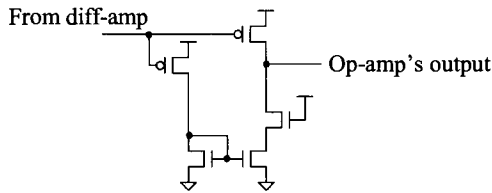


Figure 26.32 Adding a device to allow the output voltage to swing.

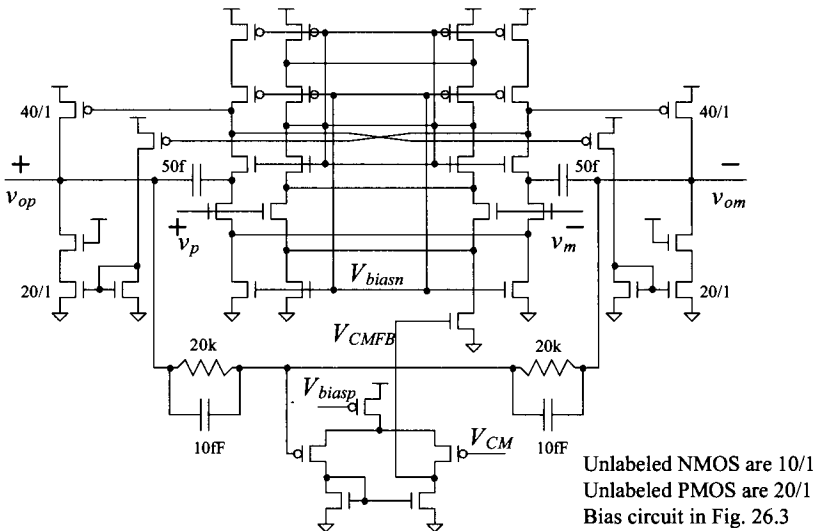


Figure 26.33 Op-amp with modified output buffer.

Simulation Results

Figures 26.34 to 26.36 show simulation results based on the circuit topologies in Figs. 26.23 to 26.25, respectively, using the op-amp in Fig. 26.33. In Fig. 26.34 we see that the outputs now swing close to the power supply rails (unlike what we saw in Fig. 26.30). Also, the gain is higher. Figure 26.35 shows the step response of the amplifier in the topology seen in Fig. 26.24. Finally, Fig. 26.36 shows the outputs of the sample-and-hold in Fig. 26.25 using the op-amp in Fig. 26.33.

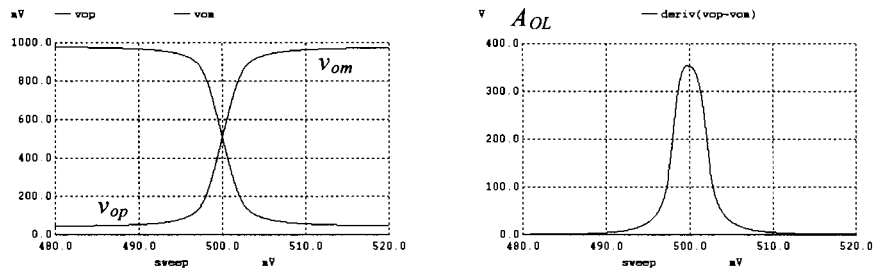


Figure 26.34 Resimulating the op-amp in Fig. 26.33 in the configuration seen in Fig. 26.23.

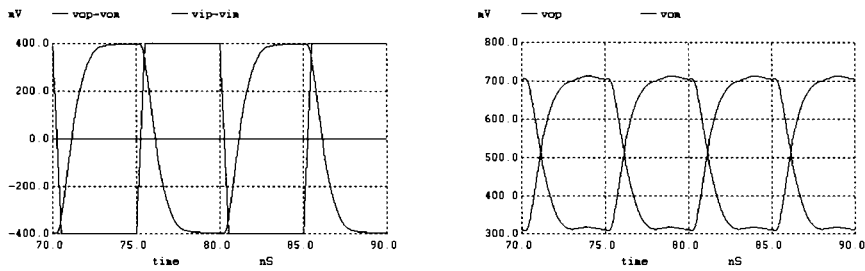


Figure 26.35 Regenerating the simulation results using the topology in Fig. 26.24 with the op-amp in Fig. 26.33.

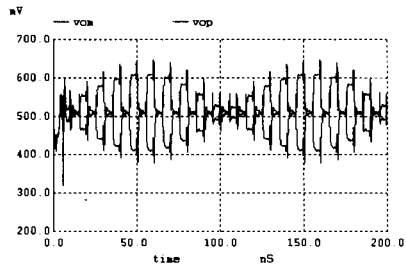


Figure 26.36 Using the op-amp in Fig. 26.33 in the sample-and-hold circuit of Fig. 26.25. Figure shows the outputs of the op-amp.

Using MOSFETs Operating in the Triode Region

For the output buffer in Fig. 26.32, we added a transistor in the drain portion of the circuit. We might wonder if we can accomplish better control of the current flowing in the output buffer by adding circuitry to the source side of the buffer. Towards answering this question, consider the portion of the output buffer seen in Fig. 26.37. If both outputs are at V_{CM} , the gate-source voltages of M2 and M3 are the same and so are the gate-source voltages of M1 and M4. If one output goes high and the other output goes low (and the outputs are centered around V_{CM}), then the net drain current of each M3 is constant. If one output goes below the threshold voltage of an NMOS device, then the balancing stops working. The practical problem of using triode-operating MOSFETs (in a CMFB circuit or any type of amplifying configuration) is that the gain through a MOSFET operating in the triode region is low (so it's difficult to provide control).

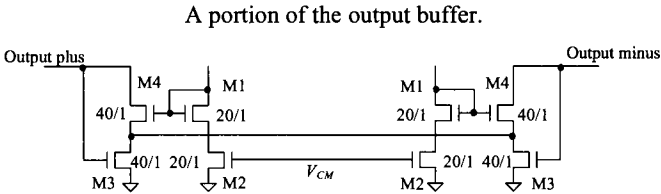


Figure 26.37 Using triode-operating MOSFETs to balance the outputs (bad).

Start-up Problems

Examine the circuit in Fig. 26.38 using the op-amp in Fig. 26.33. Because the op-amp's inputs are at 0 V, the diff-amps on the input of the op-amp are off. This causes the gates of the PMOS devices in the output buffer to be pulled to V_{DD} . The circuit remains in this state and doesn't move the outputs or the inputs up to V_{CM} . To avoid this start-up problem, we need to ensure that there is some DC path to V_{DD} or V_{CM} to "start-up" the op-amp.

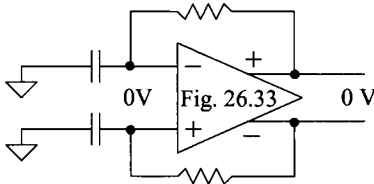


Figure 26.38 The op-amp in Fig. 26.33 won't turn on in this topology.

Lowering Input Capacitance

Consider connecting the gates of the NMOS diff-amp used for biasing to V_{CM} (see the bold line in Fig. 26.39). This reduces the input capacitance of the op-amp and shouldn't affect the normal operation as long as the common-mode voltage of the op-amp (the gates